Sathi

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Welcome back

Pick up where you left off.

CONTINUE · AI ENGINEERING · MODULE 3

Attention mechanisms

Lesson 4 of 7. Ten minutes on scaled dot-product attention, then a short check.

Continue lesson

Revision due today

5 cards · about 6 minutes

Start revision

Your mastery

Gradient descent	82%
Embeddings	68%
Backpropagation	42% · needs work
Attention	35% · needs work

Study next

Backpropagation

Weakest concept. Three recent misses on the chain rule.

Attention quiz

Check your understanding after today's lesson.

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Attention mechanisms

Beginner | Intermediate | Advanced

Explain | Give a hint | Quiz me

Why do we divide by the square root of d in attention?

Dot products grow as vectors get longer, which pushes softmax toward one-hot outputs and tiny gradients. Dividing by the square root of d keeps the scores in a stable range [1].

Try it: if each entry has variance 1, what is the variance of a dot product of length d? [2]

Simpler | Harder | Helpful | Not helpful

Ask anything about this lesson

Search the web

Send

Sources used

1 · Lecture 3 notes.pdf
Page 12

2 · Attention basics
Course lesson 2

Answers cite your course material and your own uploaded notes only.

Backpropagation

Which statement about the chain rule in backpropagation is correct?

A. Gradients are computed from the input forward

B. Each layer multiplies the incoming gradient by its local derivative · Correct answer

C. Weights are updated before gradients exist · Your answer

D. Only the final layer receives a gradient

Why B

Backpropagation moves from the output backward. Each layer takes the gradient arriving from above and multiplies it by its own local derivative.

Backpropagation mastery 42% to 48%

Next question

Ask the tutor

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My notes

Only you can see these files. The tutor uses them to answer your questions.

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Backprop cheatsheet.md — Creating search index · 70%

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Retry